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Externalities

Chapter 10

Objectives

- To understand positive and negative externalities
 - What they mean?
 - How they affect market quantity and socially beneficial quantity?
 - What is Coase Theorem?

Market Supply and Demand

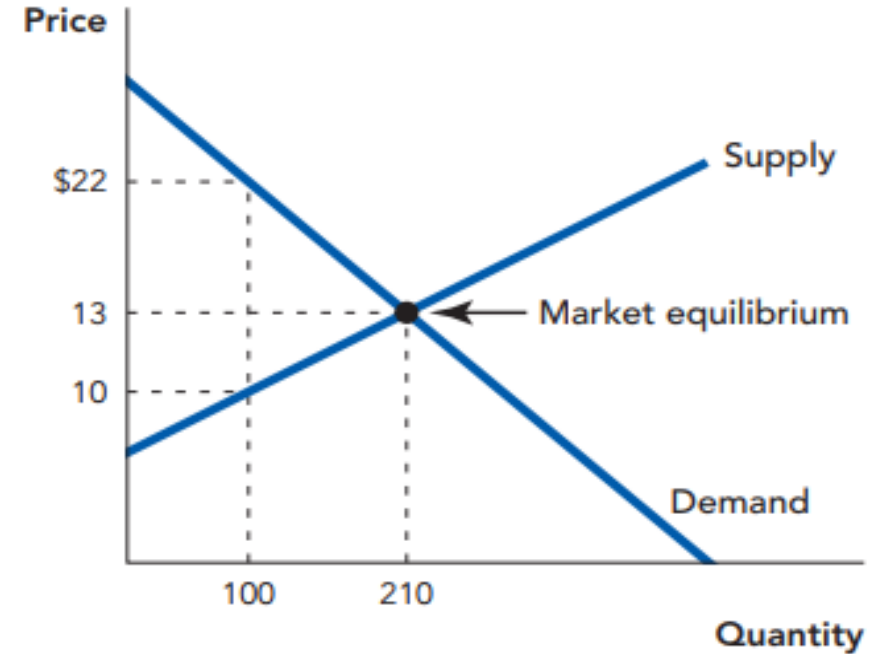
- *Typically*, well-functioning markets allocate the **sale** of goods to the buyers who value them the most.
- *Typically*, well-functioning markets allocate the **production** of goods to those who can produce the goods at the least cost.
- *Well-functioning* markets produce goods and services efficiently – and lead to **efficient outcomes**.

Review of Gains from Trade

At the 210th unit:

- The value to buyers is \$13 and the cost to sellers of producing that additional unit is \$13, so there are no further incentives to trade.
- If any fewer units were traded, gains from trade would be left on the table.
- If any more units were traded, the cost of those units would exceed their value.

FIGURE 10.1



Reviewing Gains from Trade The value of the 100th unit to buyers is \$22. The cost of the 100th unit to sellers is \$10. At the 100th unit, there is a \$12 gain from exchange. Gains from trade are maximized when a total of 210 units are exchanged. Notice that the value of the 210th unit is just equal to the cost of the 210th unit.

Market Supply and Demand

- Efficient outcomes arise because there are gains from trade and in these transactions the sellers and buyers typically incur (all) the **costs** and **benefits** from the transaction.
- Sometimes, however, there are some **costs** (or **benefits**) that are incurred by people **not involved** in the transaction.
 - **External costs** – the costs that fall on bystanders not involved in the market transaction.
 - **External benefits** – the benefits that fall on bystanders not involved in the market transaction.
 - **Externalities** are the external costs and benefits that fall on **bystanders**.

An example: What is pollution?

- What we call “pollution” is not merely “waste”, but situations where actors foist some of the costs of their waste on others



What is pollution (cont.)?

- Pollution is not simply waste.
- Pollution is waste imposed on non-consenting parties



Vaccinations

- When I get a flu shot, I am not the only one that benefits –
 - Because I received the flu shot and am less likely to get the flu, others, that I come across, are less likely to get the flu (positive externality).
 - For example, one research study finds that for every **two** flu vaccinations it prevents **one** other person from getting the flu and for every **4,000** flu vaccinations **one** life is saved.
- When I consider the costs and benefits of getting the flu vaccine, I weigh
 - the **benefits** of me not getting the flu and **not spreading** it to my family and friends
 - the **costs** of getting the flu shot in terms of **time, money**, and some **minor inconveniences**.
 - I typically do not **fully weigh** all the benefits to society when I get a flu shot.

Externalities

- Externalities arise when the **private costs** are not equal to total **social costs**.
- Externalities arise when the **private benefits** are not equal to total **social benefits**.
- When there are **negative externalities**, the market produces **too much** of the good.
- When there are **positive externalities**, the market produces **too little** of the good.

Externalities

- An **externality** arises when a person engages in an activity that influences the well being of a *bystander* and the bystander neither pays nor receives compensation.

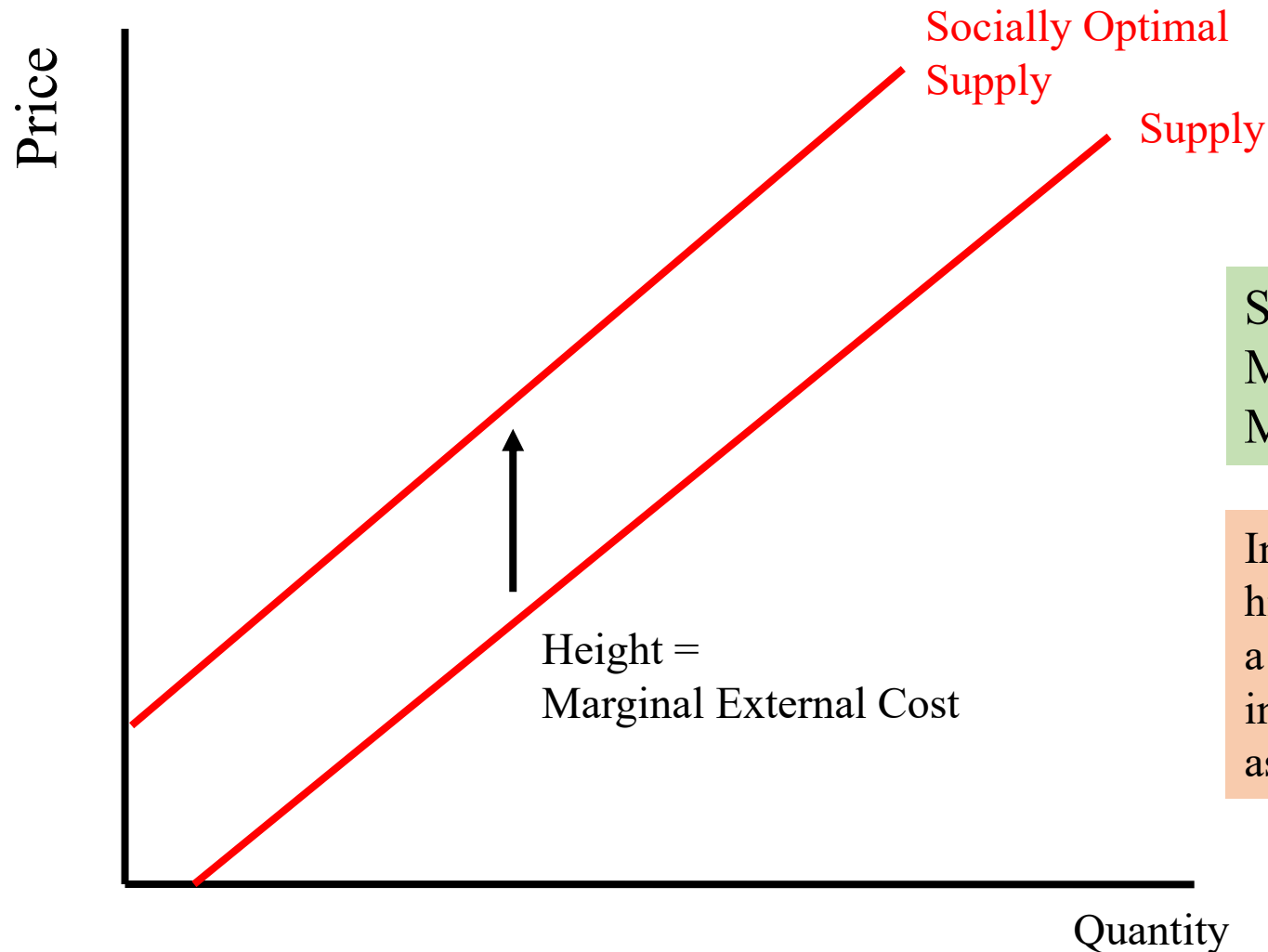
Examples of Negative Externalities

- A negative externality exists when the production (or consumption) of a results in costs imposed on society that are not explicitly or implicitly paid for by the consumer (or producer)

Examples:

- A business that pollutes the air or water.
- If you take antibiotics too often, a resistant bacterial strain is more likely to form and be less effective for others.
- If I smoke near you, you may be bothered by the second-hand smoke.
- If I play my music too loud, it may bother you.
- If I am trolling on social media websites, I make it less enjoyable for others.

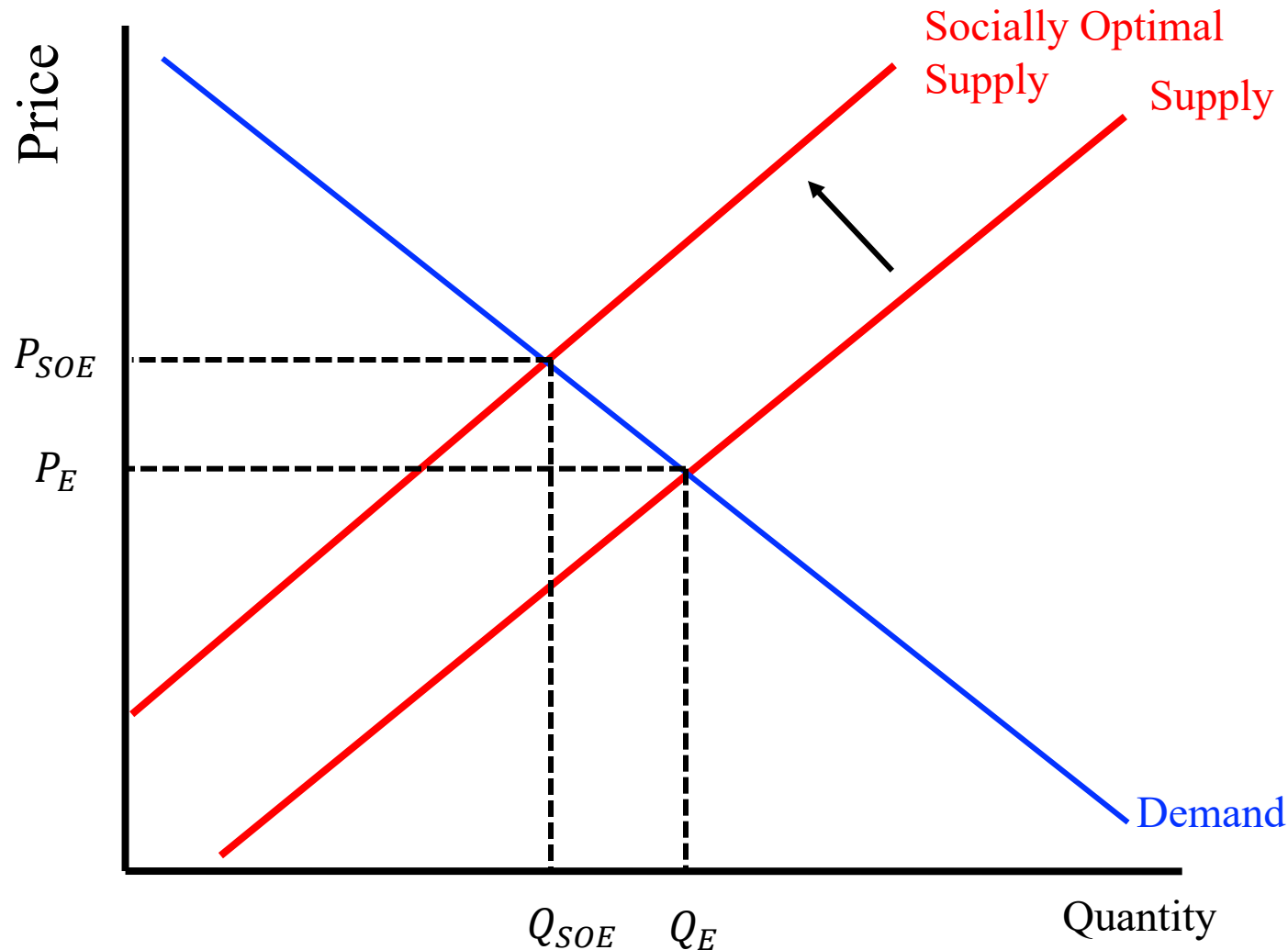
Negative Externality and External Cost (Example: Pollution)



Socially Optimal Supply =
Marginal Private Cost (Supply Curve) +
Marginal External Benefit

Intuitively: A negative externality imposes higher costs than what is simply paid by the firm. a firm pays marginal cost of labor, capital and intermediate inputs but may not include costs associated with environmental damage

Negative Externality and External Cost (Example: Pollution)



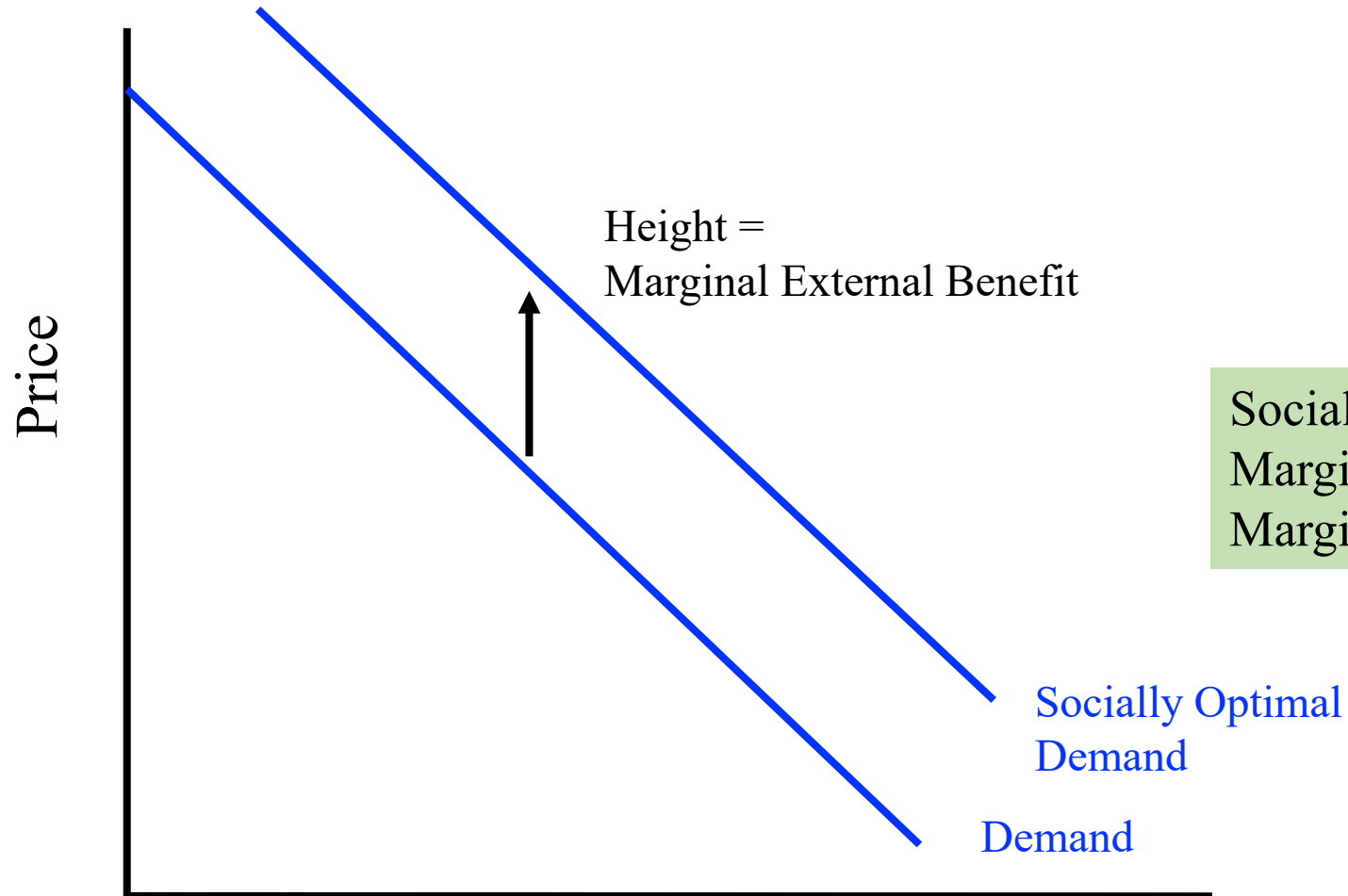
A socially optimal outcome would equate the Socially Optimal Supply with Demand.

A socially optimal outcome would correspond to a lower amount consumed and a higher price.

Positive Externality

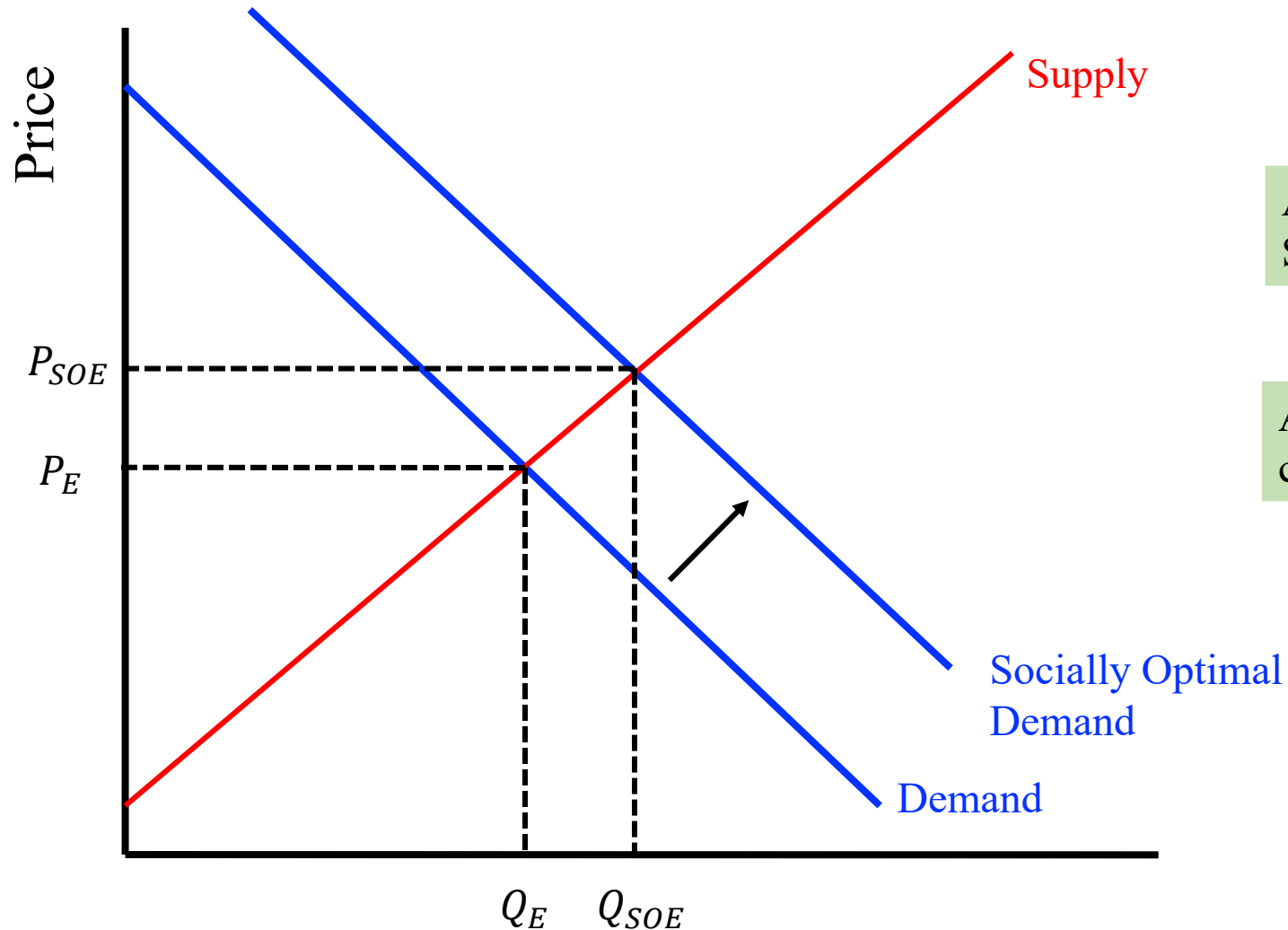
- A positive externality exists when the production (or consumption) of a provides benefits to society that are not explicitly or implicitly paid for by the consumer (or producer)
- EXAMPLES:
 - Vaccinations
 - Beekeeper and Flower gardens
 - Well kept yards / public parks
 - Innovations / Ideas

Positive Externality and External Benefit(Example: Vaccination)



Socially Optimal Demand =
Marginal Private Benefit (Demand Curve) +
Marginal External Benefit

Positive Externality and External Benefit(Example: Vaccination)

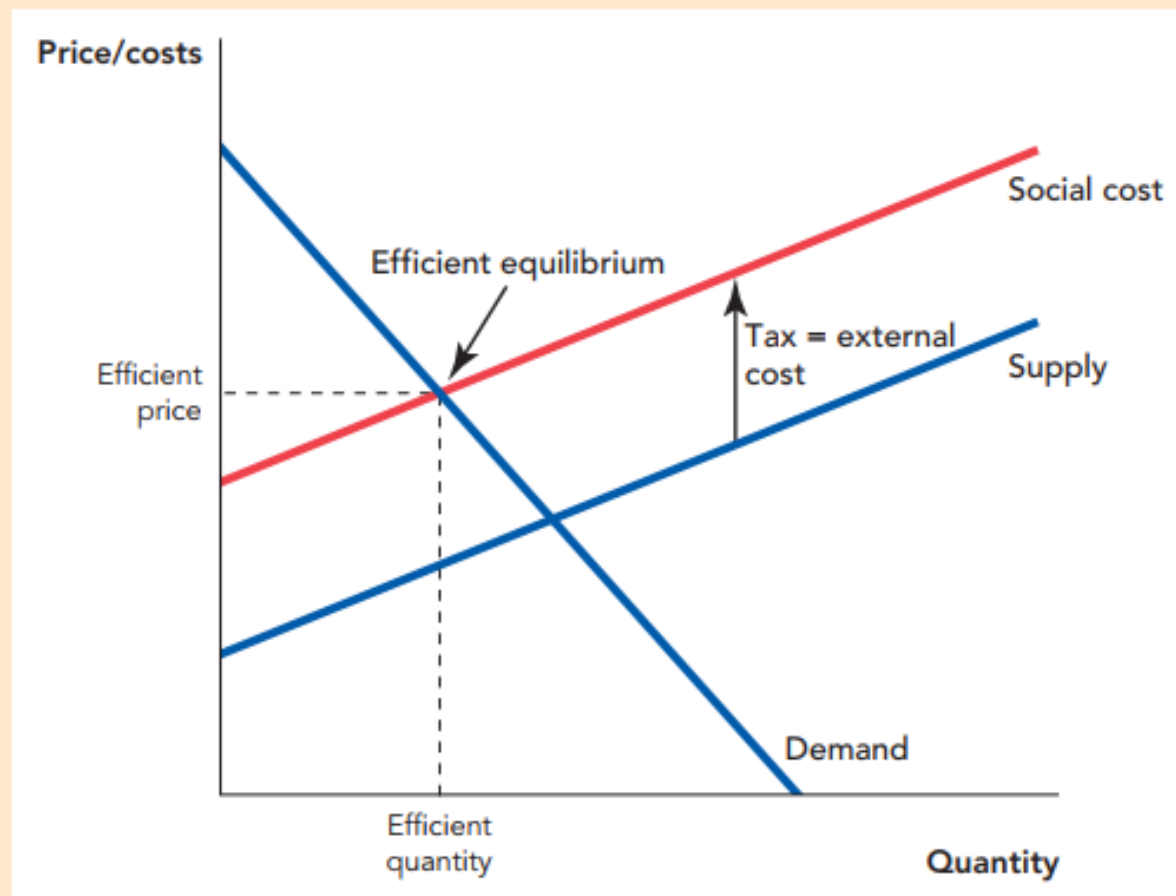


A socially optimal outcome would equate the Socially Optimal Demand and Supply

A socially optimal outcome would correspond to a higher amount consumed.

Pigouvian Tax or Subsidy

FIGURE 10.5



Comparing Tradable Allowances and Pigouvian Taxes If we knew the exact positions of the supply and demand curves, then we could always use tradable allowances to hit the efficient quantity or a tax to hit the efficient price and the equilibrium would be identical.

A subsidy on a good with an external benefit is often called a Pigouvian subsidy.

A tax on a good with an external cost is often called a Pigouvian tax.

Practice Questions

If you fail to take into account the true joy that your singing in the shower gives to the neighbors

- a. you sing too much.
- b. you sing too little.
- c. you have a comparative advantage in singing.
- d. there is a negative externality.

Practice Questions

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Practice Questions

If after many arguments, I agree not to play loud music after 10 p.m. in return for your washing my car each week

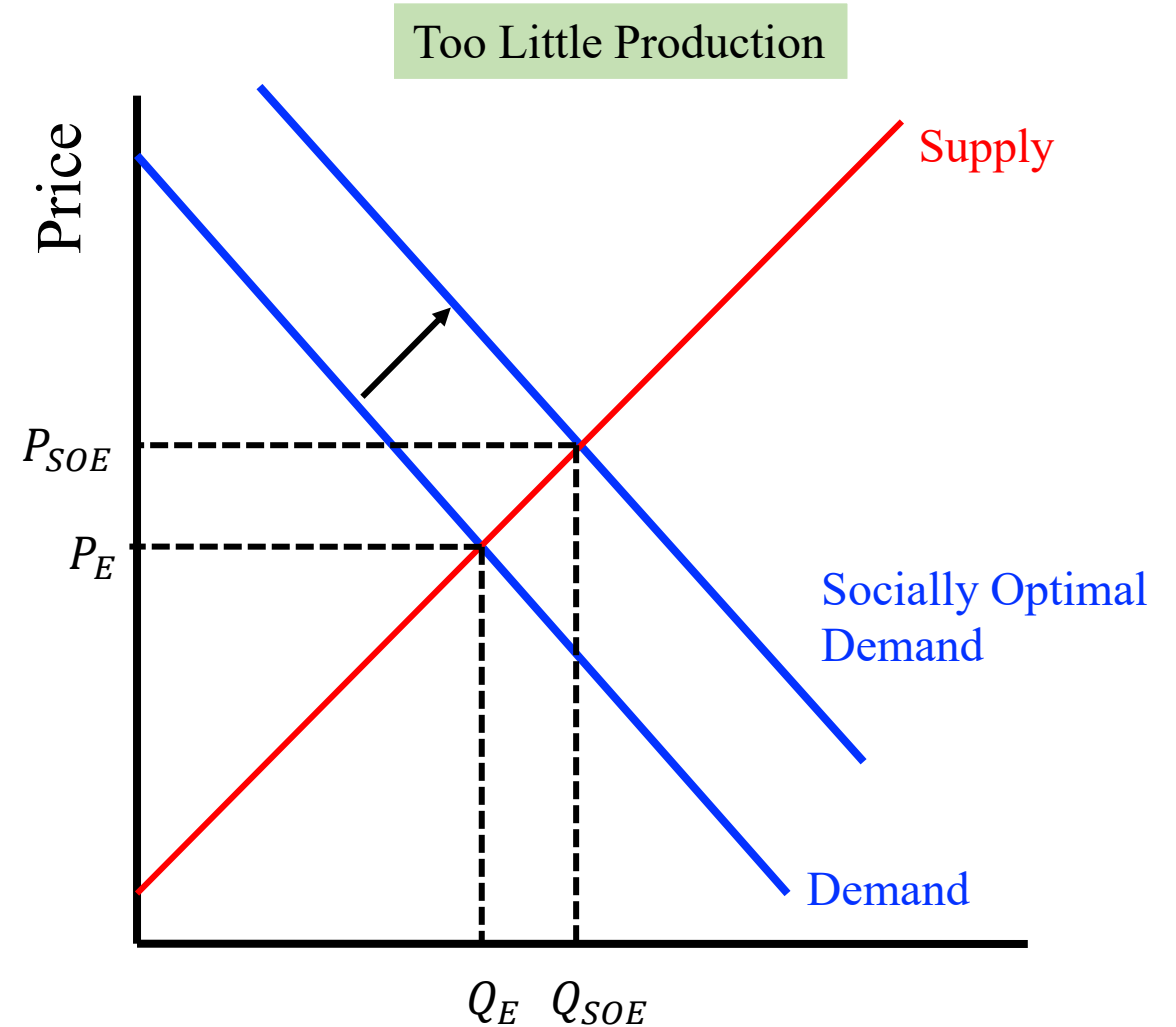
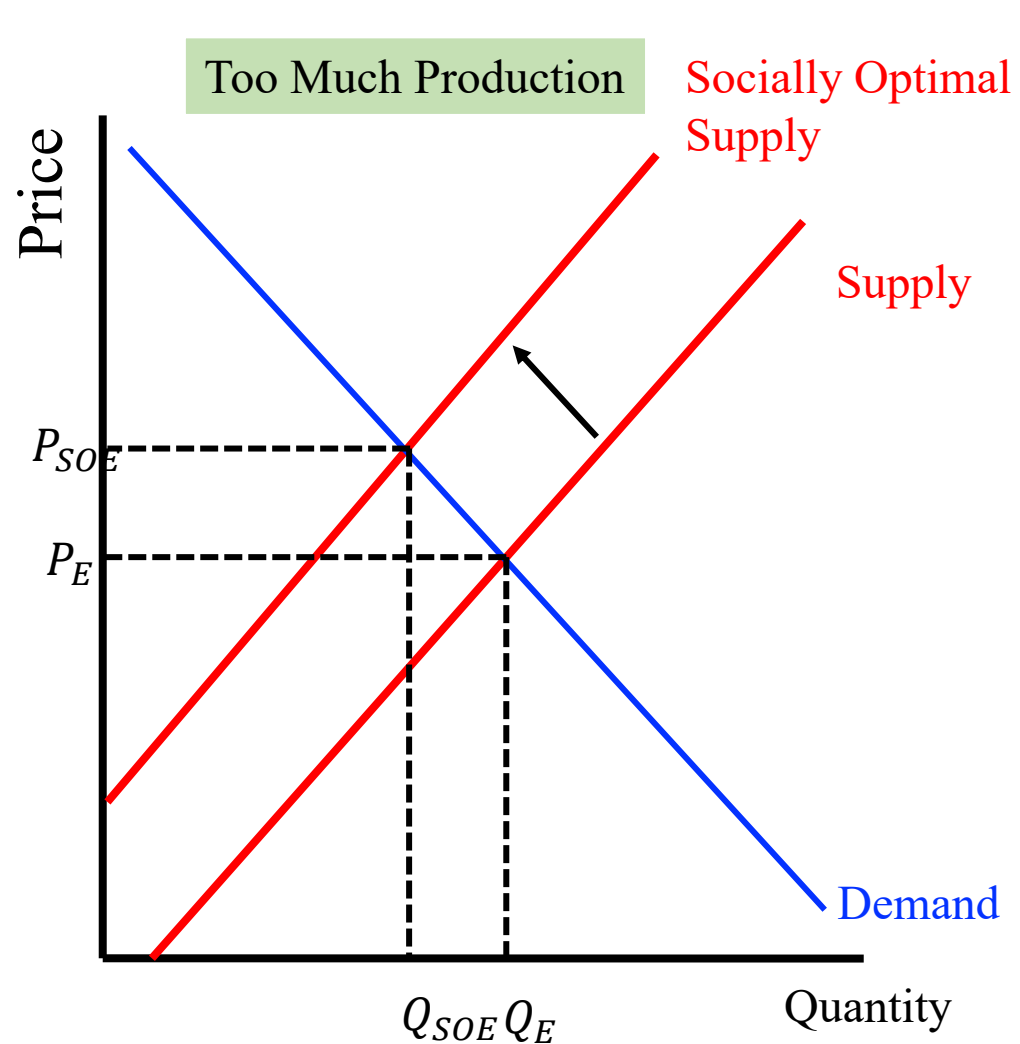
- a. my music playing has created a positive externality.
- b. your car washing creates a positive externality.
- c. there are no longer any externalities (the externality has been “internalized”).
- d. the positive and negative externalities balance exactly.

Practice Questions

If after many arguments, I agree not to play loud music after 10 p.m. in return for your washing my car each week

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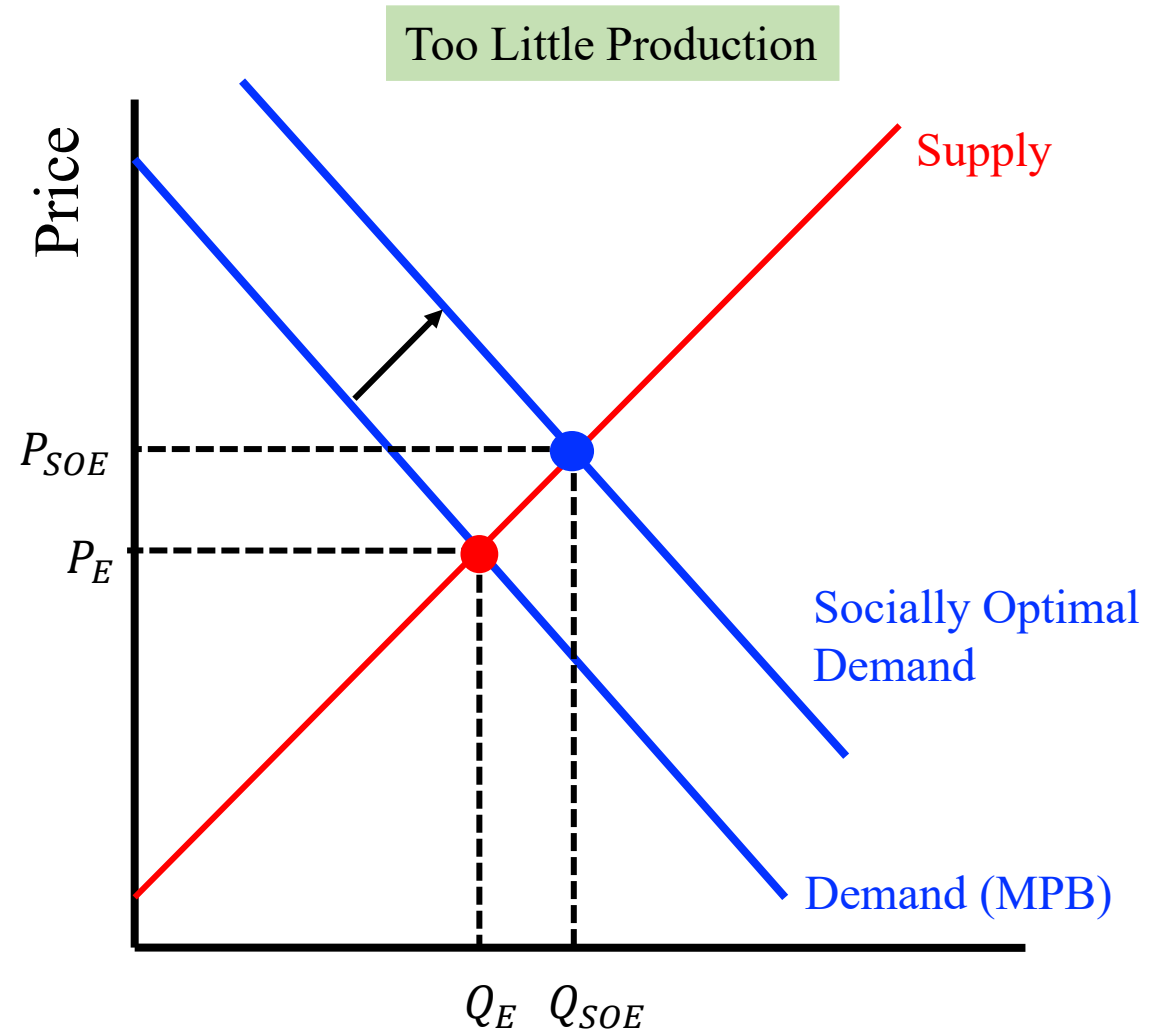
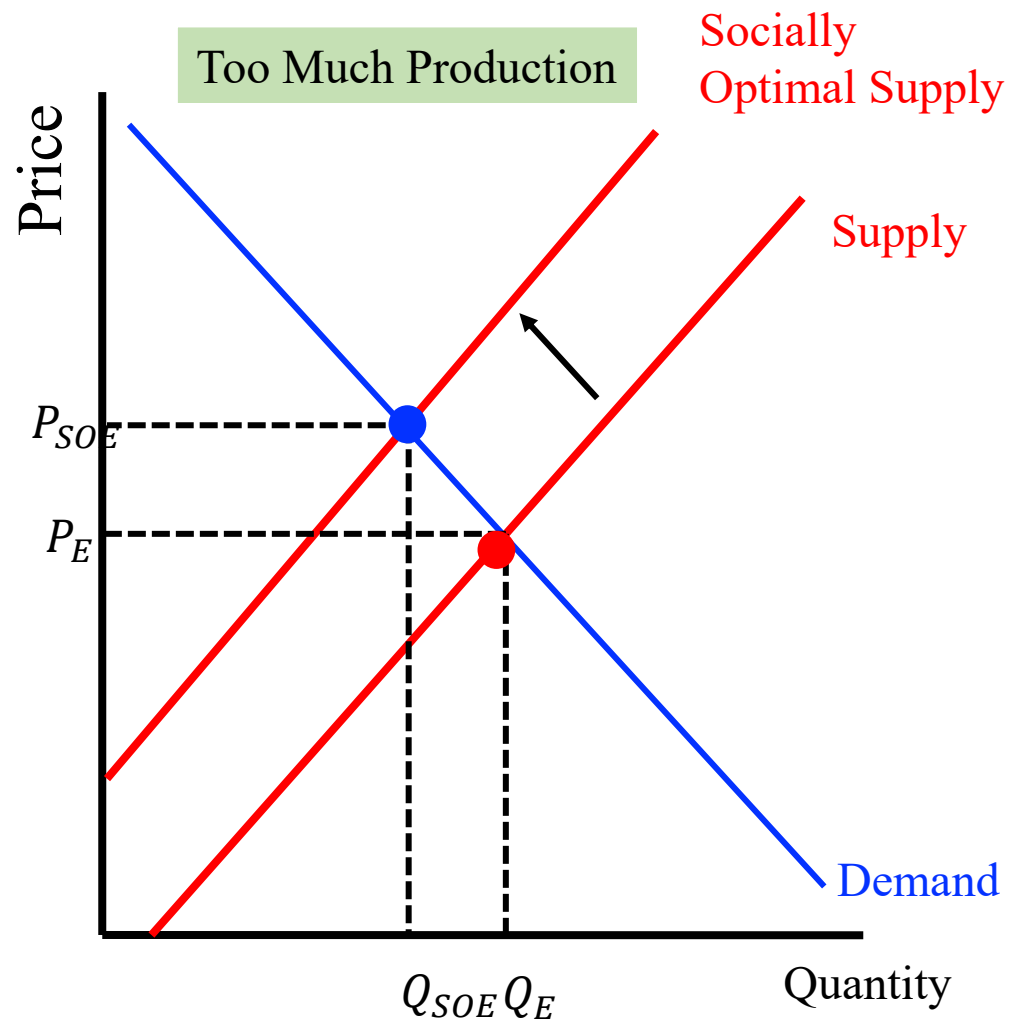
Solutions to Externality Problem



Solutions to Externality Problem

- The **socially optimal outcome** in the presence of externalities is to
 - Produce less of an item if its marginal social cost is greater than the marginal social benefit.
 - Produce more of an item if its marginal social benefit is greater than the marginal social cost.
 - The socially optimal amount of production would produce where the marginal social benefit equals the marginal social cost.

Solutions to Externality Problem

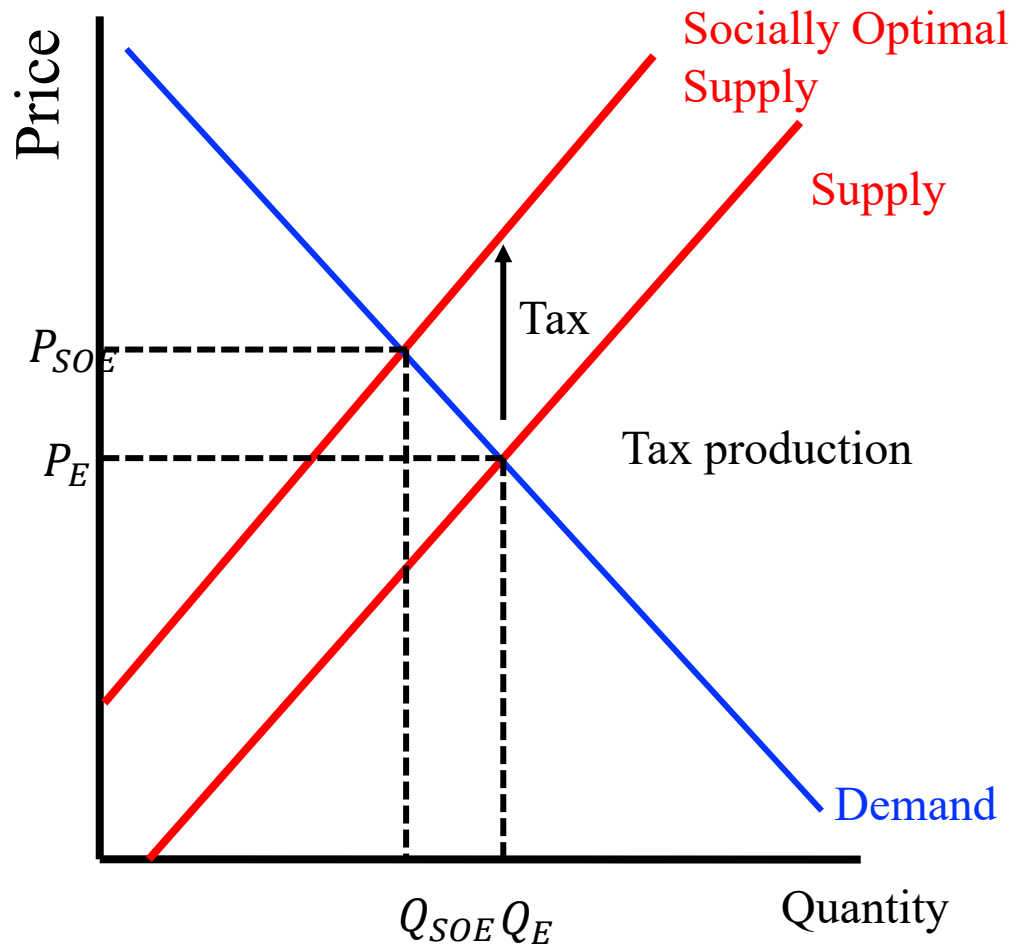


Solutions to the Externality Problem,

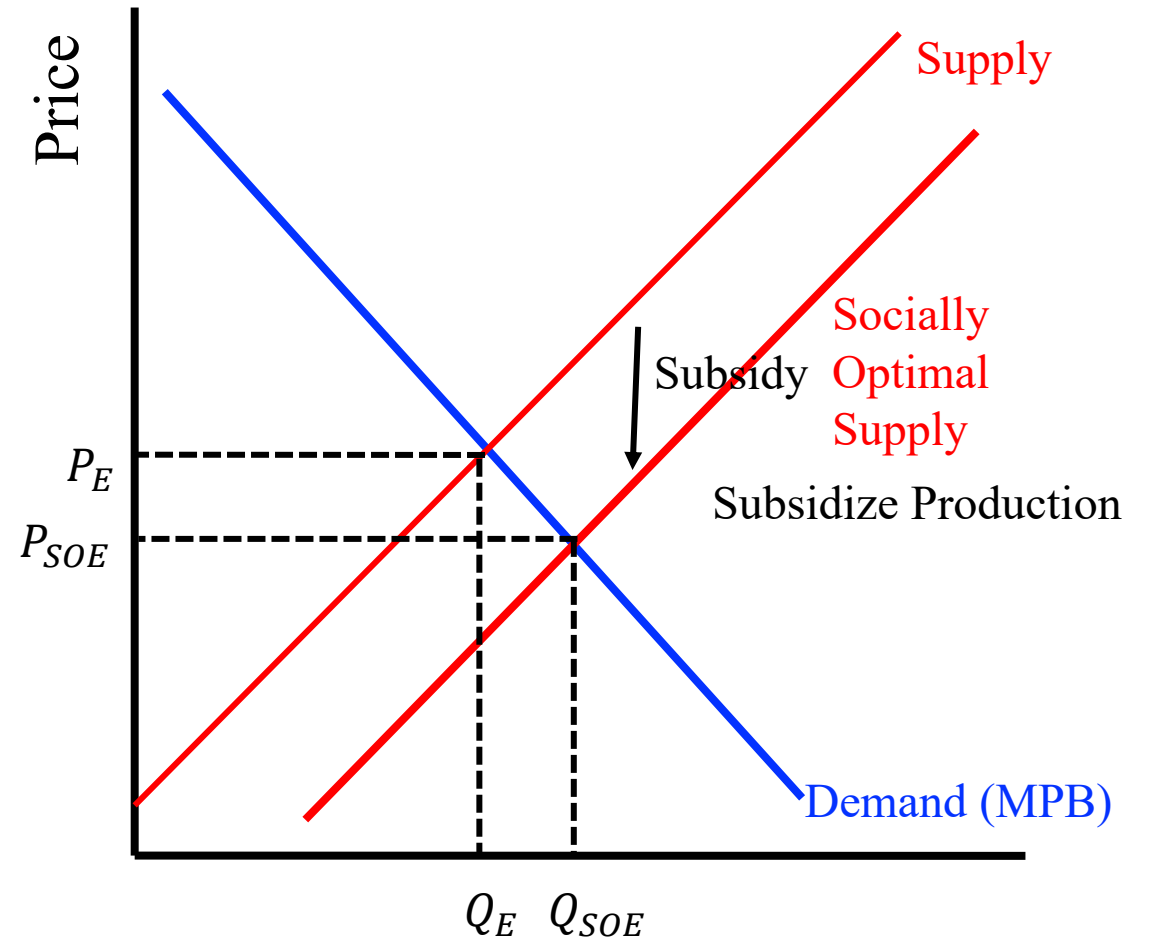
1. Fix the price: Taxes and Subsidies
2. Fix the quantity: Cap and Trade
3. Private Bargaining
4. Laws Rules and Regulations

Taxes and Subsidies

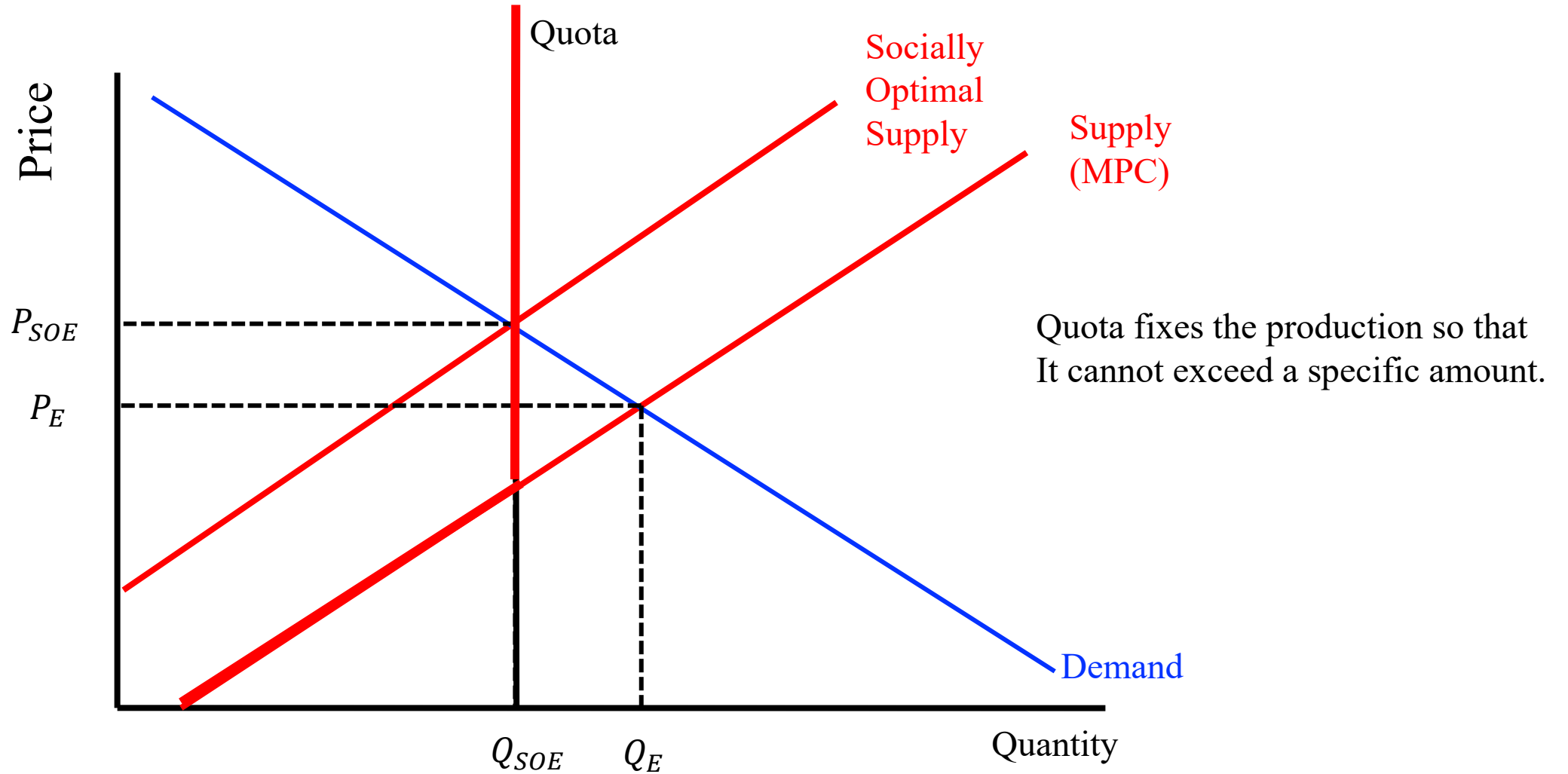
Negative Externality: Too Much Production



Positive Externality: Too Little Production



Quota



Cap and Trade

- Taxes and subsidies work well if we know (can measure) external costs and we can impose the correct tax.
- If we do not know the costs but we do have an idea of what the socially optimal quantity, then we can use quantity restrictions (quota).
- When using a quota, the government can tell each company that they can produce the same amount.
 - Government probably does not have enough information to identify which suppliers are efficient and which are less efficient.
- Cap and Trade: Government issues permits to the firms and allows them to use the permits or trade/sell to other companies.

Cap and Trade: Outcome

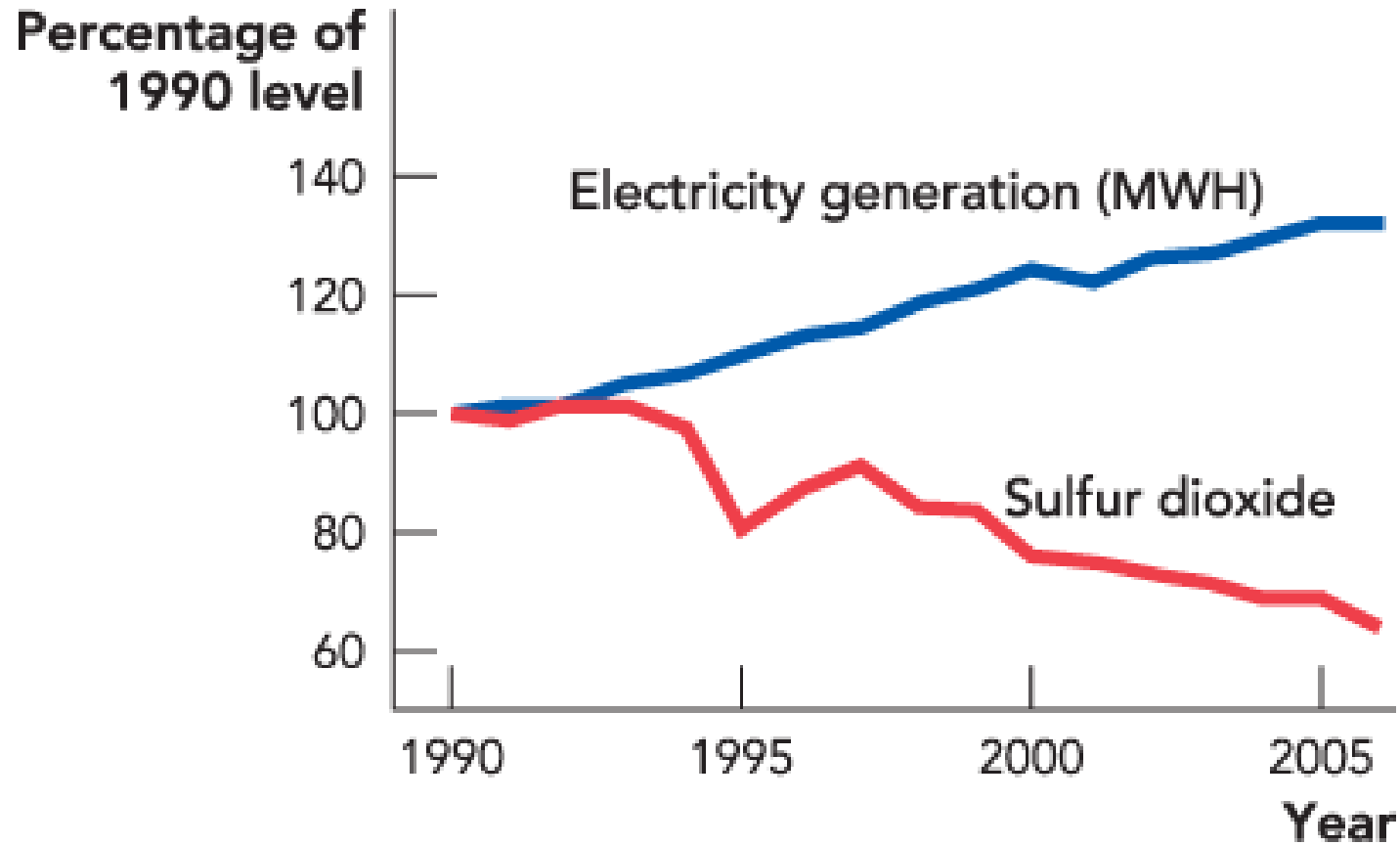


Figure 10.4



Private Bargaining

- Coase Theorem: If bargaining is costless (or low) and property rights are clearly established and enforced, then externality problem can be solved by private bargaining.

An example

- Imagine there's 2 companies that use the lake
- The meatpacker dumps waste into the lake, killing the fish
 - Is this a negative externality?
 - Moreover, what is the efficient allocation?



A meatpacking plant

Value from using the lake: \$5,000

A lake



A fishing company

Value from using the lake: \$10,000

Ownership and efficient use



A meatpacking plant

Value from using the lake: \$5,000

The meatpacker is willing to not dump into the lake if they are paid more than \$5,000

Suppose the meatpacker owns the lake

A lake



A fishing company

Value from using the lake: \$10,000

The fishing company would be willing to pay the meatpacker up to \$10,000 to not dump waste into the lake

Ownership and efficient use



A meatpacking plant

- Suppose the fishing company offers the meatpacker \$7,000 to not dump waste into the lake



A fishing company

A lake



- The opportunity cost of dumping in the lake is now \$7000. The marginal benefit is \$5000, so the meatpacker agrees to not dump.
 - The efficient allocation is achieved. There is no externality!

Ownership and efficient use



A meatpacking plant

Value from using the lake: \$5,000

A lake

The meatpacker is willing to pay up to \$5000 to be allowed to dump in the lake

Suppose the fishing company owns the lake



A fishing company

Value from using the lake: \$10,000

The fishing company would be willing to allow the meatpacker to dump if they are paid more than \$10,000

Ownership and efficient use



A meatpacking plant



A fishing company

A lake



- The meatpacker is not able to pay the fishing company enough to dump in the lake
- Therefore, the lake is used for fishing (the higher-valued use)
 - The efficient allocation is achieved. There is no externality!

“Internalizing” externalities

- Ownership ensures that the doer will face the true cost of his/her action
 - Think of waste in my basement as opposed to waste along a roadside
- In this example, it doesn't matter who owns the lake
 - As long as ownership is defined, the efficient allocation is achieved
- What matters is that once ownership is clear, all costs are taken into account

So well-defined property rights ensure nothing is external

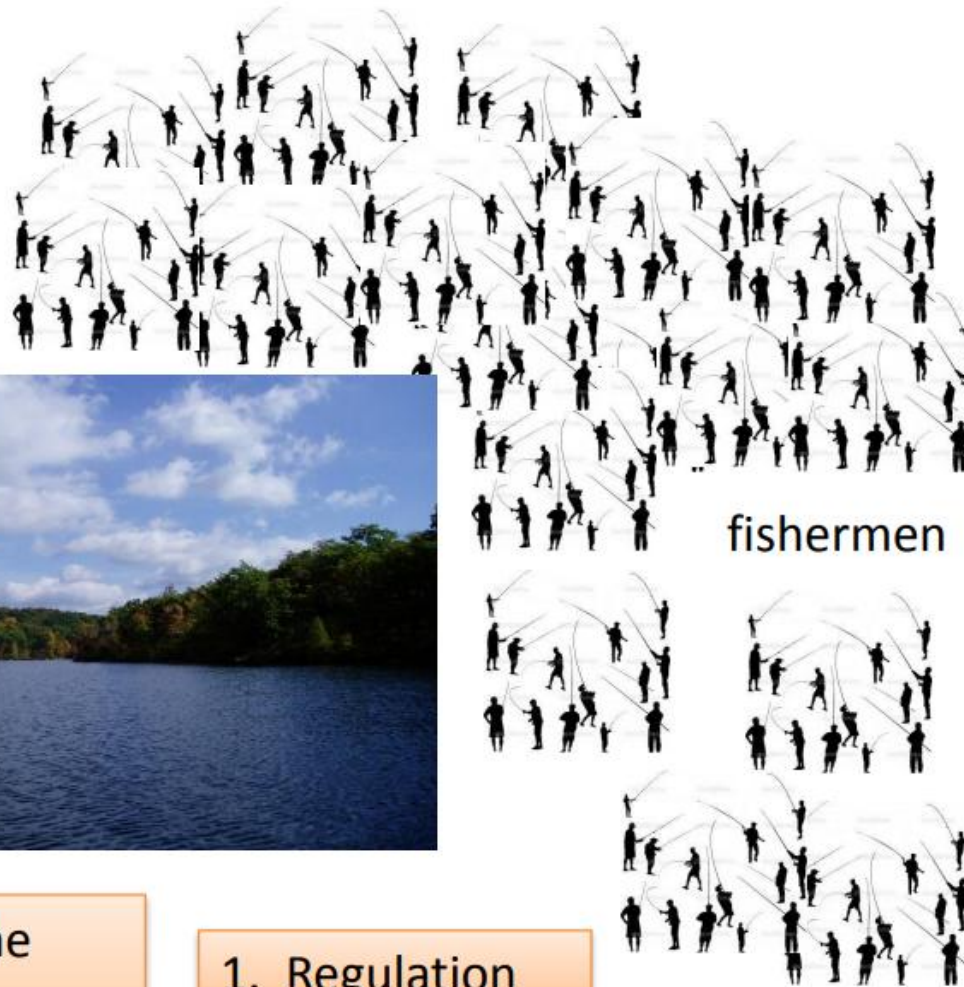


A meatpacking plant

But what if defining property rights is very costly?



A lake



Imagine now that the collective value of fishing in the lake is \$10,000. If the meatpacker owns the right to use the lake, the fishermen now have to collectively bargain with the meatpacker, which is costly. It is possible that it is too costly, and the fishermen won't be able to pay the meatpacker to stop dumping, and the efficient allocation will not be reached

Where the costs of transacting through the marketplace ("transaction costs") are very high, government policy may be preferable

1. Regulation
2. Taxation

Coase Theorem

All **costs/benefits are internalized** and there is no externality if two conditions are met.

1. Property rights are clearly defined.
2. Transaction costs are low.

Practice Questions

Externalities occur because

- a. people are short-sighted.
- b. property rights have not been clearly defined.
- c. waste is often a part of producing goods.
- d. all of the above

Practice Questions

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Practice Questions

If my neighbors, without my consent, build a garage that leaves my yard in the shade,

- a. there is a negative externality.
- b. there is property rights problem—who owns the sun?
- c. they may not be taking costs imposed on me into account
- d. all of the above. **

Practice Questions

If my neighbors, without my consent, build a garage that leaves my yard in the shade,

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Credits to @grebcomics and Cowen and Tabarrok (2014)